LungLogic Single-Lung Flow Control

PID, actuator mapping, Smith predictor, and calibration plan

Where is PID used?

Inside the flow feedback loop

What is controlled?

Measured flow tracks target flow

What is the actuator?

PPV40V proportional pinch valve

Smith predictor

Delay-compensated feedback estimate

PID calibration

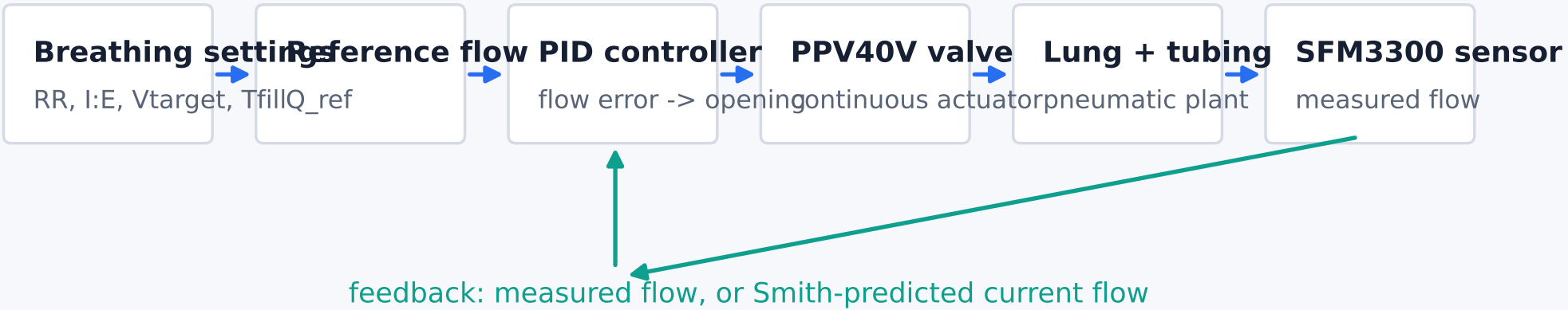
Kp, Ki, Kd, integral limit

Smith calibration

Delay, model parameters, correction gain

Main message: PID controls flow through PPV opening; Smith predictor improves the timing of the feedback signal.

Control Loop Overview



The loop is flow control. Volume is obtained by integrating flow and is used as a phase stop condition.

Where PID Is Used

- Python simulation: single_lung_pid_gui.py -> PIDController.update().
- Simulation loop calls PID during INSPIRATION and EXPIRATION.
- Hardware path: Python sends Kp, Ki, Kd and integral_limit to Arduino through CFG.
- Arduino firmware runs updatePID() at 100 Hz.
- PID is reset at state transitions to avoid carrying integral error across breathing phases.

PID law

a = clip(a_ff + Kp * e + Ki * integral(e dt) + Kd * d, 0, 1)

What PID Controls

Controlled variable

Flow rate Q measured by SFM3300
not pressure, not direct volume

Manipulated variable

PPV40V normalized opening a
range: 0 closed to 1 fully open

Inspiration reference

Q_in_ref = 0.06 * Vtarget / Tfill
250 mL, 1.30 s -> 11.54 SLM

Expiration reference

Q_out_ref = -0.06 * Vtarget / usable expiration time
PID tracks the magnitude of negative flow

Volume is a result of controlled flow: V[k] = V[k-1] + Q[k] * 1000/60 * dt.

Actuators

Continuous actuator

PPV40V proportional pinch valve
PID output controls opening

Direction actuator

Atmosphere solenoid
ON during inspiration

Direction actuator

Vacuum solenoid
ON during expiration

PPV voltage mapping

U_cmd = 5 * (1 - a)
a = 1 -> 0 V -> fully open
a = 0 -> 5 V -> fully closed

Solenoid interlock

Inspiration: ATM open, VAC closed
Expiration: ATM closed, VAC open
Dead time: both closed

The solenoids switch the pneumatic route; PID continuously modulates the pinch valve.

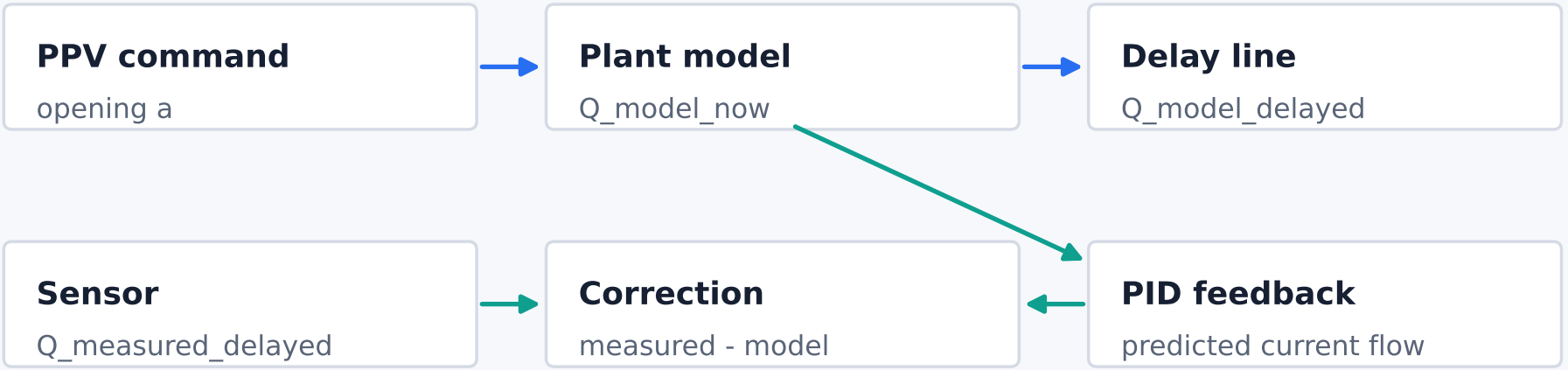
Smith Predictor

- Problem: pneumatic transport and sensor response create delayed feedback.
- Delayed feedback can make PID overshoot, oscillate, or respond conservatively.
- Smith predictor estimates the current flow using the plant model.
- The delayed real measurement is still used to correct model error.

Feedback estimate used by PID

Q_feedback = Q_model_now + gain * (Q_measured_delayed - Q_model_delayed)

Smith Predictor Position



Smith predictor changes only the feedback signal. PID law and the actuator remain the same.

Parameters Needed

PID parameters

Kp
Ki
Kd
integral_limit
control_dt

Feedforward model

capacity_ml
tau_in_s
tau_out_s
valve_exponent

Smith predictor

delay_s
correction_gain
same plant model
flow_sign

Current status

These are currently initial values for simulation and early hardware testing; they should be identified from bench data before final validation.

PID Calibration Experiments

- Open-loop PPV step tests: 20%, 40%, 60%, 80%, 100% opening for inspiration and expiration.
- Estimate maximum flow, response speed, tau_in, tau_out, and valve nonlinearity.
- Closed-loop PI tuning: set Kd = 0, increase Kp until response is fast but not oscillatory.
- Increase Ki to remove steady flow error and reduce tidal volume error.
- Validate under multiple RR, I:E ratio, target volume, and target fill time settings.

Evaluation: rise time, overshoot, steady-state flow error, tidal volume error, and cycle-to-cycle repeatability.

Smith Predictor Calibration Experiments

- Delay identification: apply a PPV opening step and measure command-to-flow response delay.
- Model alignment: compare delayed model output with measured flow after fitting tau and gamma.
- Correction gain sweep: test gain values such as 0.5, 0.75, 1.0, and 1.25.
- Smith on/off A-B test with the same PID and breathing parameters.
- Compare flow overshoot, oscillation, time to target volume, tidal volume error, and stability.

Smith predictor is useful only if delay and plant model are close enough to the real pneumatic system.

Basic Hardware Calibration Checklist

- SFM3300 zero-flow offset: all valves closed, estimate offset and noise.
- Flow direction: verify inspiration and expiration signs, then set flow_sign.
- PPV40V voltage-flow mapping: confirm 0 V fully open and 5 V fully closed.
- Inspiration step response: fit tau_in and valve exponent.
- Expiration step response: fit tau_out under regulated vacuum.
- Safety validation: emergency stop, flow limit, host watchdog, and valve interlock.

Summary

- PID is used in the flow feedback loop.
- PID controls measured flow, not pressure or direct volume.
- The continuous actuator is the PPV40V proportional pinch valve.
- Solenoid valves only select atmosphere or vacuum path.
- Smith predictor provides delay-compensated feedback to the PID.
- Calibration should identify sensor offset, flow sign, PPV mapping, model time constants, PID gains, and Smith delay.

One-sentence takeaway

State machine chooses direction, PID modulates PPV opening, Smith predictor improves feedback timing.